

a)

$$u = u(x) = x q_1 + x^2 q_2 = [x \ x^2] \begin{bmatrix} q_1 \\ q_2 \end{bmatrix} = \tilde{N}_q^T \tilde{q}$$

$$\epsilon_x = \frac{du}{dx} = B_q \tilde{q} \Rightarrow \epsilon_x = \begin{bmatrix} 1 & 2x \end{bmatrix} \begin{bmatrix} q_1 \\ q_2 \end{bmatrix}$$

① B_q

$$B_q = ? \quad \checkmark \quad k_q = EA \int_0^L B_q^T B_q dx = ? \quad \tilde{S}_q = - \int_0^L N_q^T p(x) dx = ?$$

② $k_q = EA \int_0^L \begin{bmatrix} 1 \\ 2x \end{bmatrix} \begin{bmatrix} 1 & 2x \end{bmatrix} dx = EA \int_0^L \begin{bmatrix} 1 & 2x \\ 2x & 4x^2 \end{bmatrix} dx = EA \begin{bmatrix} x & x^2 \\ x^2 & \frac{4}{3}x^3 \end{bmatrix} \Big|_0^L = EA \begin{bmatrix} L & L^2 \\ L^2 & \frac{4}{3}L^3 \end{bmatrix} = k_q$

③ $\tilde{S}_q = - \int_0^L \begin{bmatrix} x \\ x^2 \end{bmatrix} p_0 x^2 dx = -p_0 \begin{bmatrix} \frac{x^4}{4} \\ \frac{x^5}{5} \end{bmatrix} \Big|_0^L = \begin{bmatrix} -\frac{p_0 L^4}{4} \\ -\frac{p_0 L^5}{5} \end{bmatrix} = \tilde{S}_q$

$$\Rightarrow k^{-1} = \frac{3}{E^2 L^4} \begin{bmatrix} \frac{4}{3} E A L^2 & -E A L^2 \\ -E A L^2 & E A L^3 \end{bmatrix} = \frac{1}{EA} \begin{bmatrix} \frac{4}{3} & -\frac{3}{L} \\ -\frac{3}{L} & \frac{3}{L^2} \end{bmatrix} = k^{-1}$$

$EA L \cdot \frac{4}{3} E A L^2 - E A L^2 \cdot E A L^3 = \frac{E^2 L^4}{3}$

④ $k \tilde{q} = -\tilde{S}_q$

$$\tilde{q} = k^{-1} (-\tilde{S}_q) = \frac{1}{EA} \begin{bmatrix} \frac{4}{3} & -\frac{3}{L} \\ -\frac{3}{L} & \frac{3}{L^2} \end{bmatrix} \begin{bmatrix} \frac{p_0 L^4}{4} \\ \frac{p_0 L^5}{5} \end{bmatrix} = \begin{bmatrix} \frac{4}{3} \frac{p_0 L^4}{4} - \frac{3}{L^2} \frac{p_0 L^5}{5} \\ -\frac{3}{L^2} \frac{p_0 L^4}{4} + \frac{3}{L^3} \frac{p_0 L^5}{5} \end{bmatrix} = \begin{bmatrix} \frac{2 p_0 L^3}{5 EA} \\ -\frac{3 p_0 L^2}{20 EA} \end{bmatrix} = \tilde{q}$$

⑤ $u = u(x) = [x \ x^2] \begin{bmatrix} \frac{2 p_0 L^3}{5 EA} \\ -\frac{3 p_0 L^2}{20 EA} \end{bmatrix} = -\frac{3 p_0 L^2}{20 EA} x^2 + \frac{2 p_0 L^3}{5 EA} x = u(x)$

⑥ $\epsilon_x = \frac{du}{dx} = -\frac{3 p_0 L^2}{10 EA} x + \frac{2 p_0 L^3}{5 EA} \Rightarrow \sigma_x = E \epsilon_x = -\frac{3 p_0 L^2}{10 A} x + \frac{2 p_0 L^3}{5 A} = \sigma_x$

b) $u = N_1 v_1 + N_2 v_2 = [N_1 \ N_2] \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \tilde{N} \tilde{v}$

① $N_i(x) = \prod_{j=1}^{n-1} \frac{x - x_j}{x_i - x_j}$

$$N_1(x) = \prod_{j=2}^2 \frac{x - x_j}{x_1 - x_j} = \frac{x - 0}{\frac{L}{2} - 0} \cdot \frac{x - L}{\frac{L}{2} - L} = \frac{-4x(x-L)}{L^2} = \frac{4x(L-x)}{L^2} = N_1(x)$$

$$N_2(x) = \prod_{j=1}^1 \frac{x - x_j}{x_2 - x_j} = \frac{x - 0}{L - 0} \cdot \frac{x - \frac{L}{2}}{L - \frac{L}{2}} = \frac{2x(x - \frac{L}{2})}{L^2} = \frac{x(2x - L)}{L^2} = N_2(x)$$

$$\tilde{N} = \begin{bmatrix} \frac{4x(L-x)}{L^2} & \frac{x(2x-L)}{L^2} \\ \frac{-4x^2 + 4Lx}{L^2} & \frac{2x^2 - Lx}{L^2} \end{bmatrix}$$

② $\epsilon_x = \frac{du}{dx} = B \tilde{v} = \begin{bmatrix} \frac{-8x + 4L}{L^2} & \frac{4x - L}{L^2} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$

③ $k = EA \int_0^L \tilde{B}^T B dx = EA \int_0^L \begin{bmatrix} \frac{-8x + 4L}{L^2} & \frac{4x - L}{L^2} \end{bmatrix} \begin{bmatrix} \frac{-8x + 4L}{L^2} & \frac{4x - L}{L^2} \end{bmatrix} dx = EA \int_0^L \begin{bmatrix} \frac{(4L - 8x)^2}{L^4} & \frac{(4x - L)(4L - 8x)}{L^4} \\ \frac{(4x - L)(4L - 8x)}{L^4} & \frac{(4x - L)^2}{L^4} \end{bmatrix} dx$

④ $\tilde{S} = - \int_0^L \tilde{N}^T p(x) dx = - \int_0^L \begin{bmatrix} \frac{-4x^2 + 4Lx}{L^2} \\ \frac{2x^2 - Lx}{L^2} \end{bmatrix} p_0 x^2 dx = - \int_0^L \begin{bmatrix} \frac{-4 p_0 x^4 + 4 p_0 L x^3}{L^2} \\ \frac{2 p_0 x^3 - p_0 L x^2}{L^2} \end{bmatrix} dx$

$$= \frac{EA}{3L} \begin{bmatrix} 16 & -8 \\ -8 & 7 \end{bmatrix} = k$$

$$= - \int_0^L \begin{bmatrix} \frac{4 p_0 x^5 + p_0 L x^4}{L^2} \\ \frac{2 p_0 x^5 - \frac{p_0}{4} L x^4}{L^2} \end{bmatrix} dx = \begin{bmatrix} \frac{-p_0 L^3}{5} \\ \frac{-3 p_0 L^3}{20} \end{bmatrix} = \tilde{S}$$

$$-\frac{4}{5} + \frac{5}{5} = \frac{1}{5} p_0 L^5$$

$$\frac{8}{20} - \frac{5}{5} = \frac{1}{5} p_0 L^5$$

NR: $\int_0^L \frac{(4L - 8x)^2}{L^4} dx = \frac{16L^2 x - 32Lx^2 + \frac{64}{3}x^3}{L^4} \Big|_0^L = \frac{16L^2 - 32L^2 + \frac{64}{3}L^2}{L^4} = \frac{16}{3L} = k_{11}$

$$\int_0^L \frac{(4x - L)(4L - 8x)}{L^4} dx = \frac{-\frac{32}{3}x^3 + 8x^2L + 4x^2L - 4L^2x}{L^4} \Big|_0^L = \frac{-\frac{32}{3}L^3 + \frac{24}{3}L^3 + \frac{12}{3}L^3 - \frac{4}{3}L^3}{L^4} = \frac{-8}{3L} = k_{12}$$

$$\int_0^L \frac{(4x - L)^2}{L^4} dx = \frac{\frac{16}{3}x^3 - 4x^2L + L^2x}{L^4} \Big|_0^L = \frac{\frac{16}{3}L^3 - \frac{4}{3}L^3 + \frac{3}{3}L^3}{L^4} = \frac{7}{3L} = k_{22}$$



⑥ $\underline{k} \underline{\vec{v}} = -\underline{\vec{S}}^m$

$$\frac{EA}{3L} \begin{bmatrix} 16 & -8 \\ -8 & 7 \end{bmatrix} \cdot \underline{\vec{v}} = \begin{bmatrix} -\frac{P_0 L^3}{5} \\ -\frac{3P_0 L^3}{20} \end{bmatrix}$$

$$\underline{k}^{-1} = \frac{3L^2}{16EA} \begin{bmatrix} 7EA & 8EA \\ 8EA & 16EA \end{bmatrix} = \frac{L}{16EA} \begin{bmatrix} 7 & 8 \\ 8 & 16 \end{bmatrix}$$

$$\underline{\vec{v}} = \underline{k}^{-1} \underline{\vec{S}}^m = \frac{L}{16EA} \begin{bmatrix} 7 & 8 \\ 8 & 16 \end{bmatrix} \begin{bmatrix} \frac{P_0 L^3}{5} \\ \frac{3P_0 L^3}{20} \end{bmatrix} = \begin{bmatrix} \frac{7L}{16EA} \frac{P_0 L^3}{5} + \frac{8L}{16EA} \frac{3P_0 L^3}{20} \\ \frac{8L}{16EA} \frac{P_0 L^3}{5} + \frac{16L}{16EA} \frac{3P_0 L^3}{20} \end{bmatrix} = \begin{bmatrix} \frac{7P_0 L^4}{80EA} + \frac{24P_0 L^4}{320EA} \\ \frac{8P_0 L^4}{80EA} + \frac{48P_0 L^4}{320EA} \end{bmatrix} = \begin{bmatrix} \frac{13P_0 L^4}{80EA} \\ \frac{P_0 L^4}{4EA} \end{bmatrix} = \underline{\vec{v}}$$

⑦ $u(x) = \underline{N} \underline{\vec{v}} = \begin{bmatrix} -\frac{4x^2+4Lx}{L^2} & \frac{2x^2-Lx}{L^2} \end{bmatrix} \frac{P_0 L^4}{EA} \begin{bmatrix} \frac{13}{80} \\ \frac{1}{4} \end{bmatrix} = \frac{P_0 L^2}{EA} \left(\frac{-52x^2+52Lx}{80} + \frac{2x^2-Lx}{4} \right) = \frac{P_0 L^2}{20EA} (-3x^2+8Lx) = u(x)$

⑧ $\epsilon = \frac{du}{dx} = \frac{P_0 L^2}{20EA} (-6x+8L) = \frac{P_0 L^2}{10EA} (-3x+4L)$

$\sigma = \epsilon \cdot E = \frac{P_0 L^2}{10A} (-3x+4L) = \sigma(x)$

c) $\underline{\vec{v}} = \underline{A} \underline{\vec{q}} \quad u(x) = \underline{N}_1 \underline{\vec{q}} = [x \ x^2] \begin{bmatrix} q_1 \\ q_2 \end{bmatrix}$

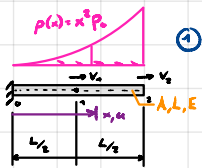
① $\left. \begin{array}{l} v_1 \hat{=} \frac{1}{2} \Rightarrow v_1 = u(\frac{1}{2}) = \frac{1}{2} q_1 + \frac{L^2}{4} q_2 \\ v_2 \hat{=} L \Rightarrow v_2 = u(L) = L q_1 + L^2 q_2 \end{array} \right\} \underline{\vec{v}} = \underbrace{\begin{bmatrix} \frac{1}{2} & \frac{L^2}{4} \\ L & L^2 \end{bmatrix}}_{\underline{A}} \underline{\vec{q}}$

② $\underline{k} = \underline{A}^T \underline{k}_q \underline{A}^{-1}$

$$\underline{A}^{-1} = \frac{4}{L^3} \begin{bmatrix} L^2 & -\frac{L^2}{4} \\ -L & \frac{L}{2} \end{bmatrix} = \begin{bmatrix} \frac{4}{L} & -\frac{1}{L} \\ -\frac{4}{L^2} & \frac{2}{L^2} \end{bmatrix}$$

$$\underline{k} = \begin{bmatrix} \frac{4}{L} & -\frac{1}{L} \\ -\frac{4}{L^2} & \frac{2}{L^2} \end{bmatrix} EA \begin{bmatrix} L & L^2 \\ L^2 & \frac{1}{3} L^3 \end{bmatrix} \begin{bmatrix} \frac{4}{L} & -\frac{1}{L} \\ -\frac{4}{L^2} & \frac{2}{L^2} \end{bmatrix} = \begin{bmatrix} \frac{16EA}{3L} & -\frac{8EA}{3L} \\ -\frac{8EA}{3L} & \frac{7EA}{3L} \end{bmatrix} = \frac{EA}{3L} \begin{bmatrix} 16 & -8 \\ -8 & 7 \end{bmatrix}$$

d)



① $\frac{d}{dx} \left(EA \frac{du}{dx} \right) + p(x) = 0$

$\Rightarrow EA \frac{d^2 u}{dx^2} + p(x) = 0$

$\Rightarrow EA u'' + P_0 x^2 = 0$

$\Rightarrow EA u'' = -P_0 x^2$

$EA u' = -\frac{P_0}{3} x^3 + C_1$

$EA u = -\frac{P_0}{12} x^4 + C_1 x + C_2$

$\Rightarrow u(x) = \frac{P_0}{12EA} (-x^4 + 4L^3 x)$

$\sigma = \epsilon \cdot E = \frac{du}{dx} \cdot E \Rightarrow N = \frac{du}{dx} \cdot EA$

1: $N(x) = \frac{du}{dx} EA = -\frac{P_0}{3} x^3 + C_1$

$N(L) = 0 = -\frac{P_0}{3} L^3 + C_1 \Rightarrow C_1 = \frac{P_0 L^3}{3}$

$\Rightarrow u' = \frac{1}{EA} \left(-\frac{P_0}{3} x^3 + \frac{P_0 L^3}{3} \right)$

$\sigma = u' E = \frac{-P_0 x^3}{3A} + \frac{P_0 L^3}{3A} = \frac{P_0}{3A} (L^3 - x^3) = \sigma(x)$

2: $EA u'' = -\frac{P_0}{12} x^4 + \frac{P_0 L^3}{3} x + C_2$

$u(L) = 0 = C_2$

$\Rightarrow C_2 = 0$

② For a quadratic load $P_0 x^2$ the analytic solution for $u(x)$ will be a 4th order polynomial.

As only quadratic interpolation functions are used, the finite element solution cannot mimic the analytic solution.

For a constant load P_0 , the analytic solution for $u(x)$ will be a 2nd order polynomial.

This matches the degree of the interpolation functions; which means the solutions will match.

